

Tecnológico de Monterrey - CCM
Advanced Artificial Intelligence for Data Science I
Assignment: Statistical and Probabilistic Analysis of Spotify Popularity

Objective: To apply fundamental concepts of descriptive statistics and probability to analyze the characteristics of the most popular songs on Spotify. Students will calculate and interpret measures of central tendency and dispersion, culminating in an analysis of relative variability (using the Coefficient of Variation) and a practical application of Bayes' Theorem.

Dataset:

- **Name:** Most Streamed Spotify Songs 2024
- **Source:** Kaggle – also available at:
<https://drive.google.com/file/d/1hUvKweiaYPyRxT5knbO1x9qmNmIFZRSW/view?usp=sharing>
- **Description:** Contains information about the most streamed songs on Spotify in 2024, including popularity variables across different platforms and the total number of streams.

Instructions:

Part 1: Descriptive Statistics Analysis (60 Points)

The goal is to describe the characteristics of musical success through a rigorous statistical analysis, emphasizing the correct interpretation of both visual and numerical metrics.

1. Data Loading and Preparation (5 pts):

- Load the dataset and perform the basic cleaning (convert relevant columns to numeric, handle nulls with the median).

2. Visualizing Data Distributions (15 pts):

- Create histograms for the **Spotify Streams** and **YouTube Views** columns.
- Interpretation:
 - Describe the shape of these histograms. **What does the "long tail" that you observe in the graphs tell you about how success is concentrated in a few songs?**

3. Measures of Central Tendency (15 pts):

- For **Spotify Streams** and **YouTube Views**, calculate the Mean and the Median.
- Interpretation:
 - Linking back to the histogram you just plotted, **explain why the mean is "pulled" to the right by the long tail, while the median remains closer to the center of the main data cluster.**

4. Measures of Dispersion and Relative Variability (25 pts):

- For **Spotify Popularity**, **Spotify Streams**, and **YouTube Views**, calculate:
 - Standard Deviation.

- Coefficient of Variation (CV).
- Interpretation: Answer in your report:
 - Using the Coefficient of Variation, **compare the variability of the three columns. Order them from most consistent to most volatile and explain what this means in the context of a song's success.**

Part 2: Probabilistic Analysis with Bayes' Theorem (40 Points)

In this section, you will investigate the relationship between a song's popularity and whether its content is explicit.

1. Defining Events (10 pts):

- Create a categorical variable to define **Event A: "The song has High Popularity"**. Define this event for all songs where **Spotify Popularity** is **greater than or equal to 85**.
- **Event B: "The song is Explicit"** is given by the **Explicit Track** column.

2. Calculating Prior Probabilities (15 pts):

- Calculate the following probabilities directly from your dataset:
 - **P(A)**: The prior probability that a song in this dataset has "High Popularity".
 - **P(B)**: The prior probability that a song is "Explicit".
 - **P(A|B)**: The conditional probability that a song has "High Popularity" **given that** it is "Explicit".

3. Applying Bayes' Theorem (15 pts):

- Using Bayes' Theorem, answer the key question:
If we randomly select a song and observe that it has "High Popularity", what is the new probability that this song is "Explicit"?
- You must calculate **P(B|A)**. Show the Bayes' Theorem formula, substitute the values you calculated, and present the result.
- Compare your final result (the posterior probability $P(B|A)$) with the initial probability (the prior probability $P(B)$). **Did knowing that a song is very popular make it more or less likely to be explicit?**

Deliverable

1. **Report (PDF/Word)**: Must contain all explanations, justifications, interpretations, and conclusions, along with code snippets and their results.
2. **Notebook (.ipynb)**: The complete and functional code for your analysis.

Evaluation

- Descriptive Statistics (60 points total)
- Probabilistic Analysis (40 points total)

Note: The grade for this assignment does not directly count toward the final evaluation. However, a minimum of 80 points is required in order to unlock the Quiz. The Quiz grade will be the one considered for the final module evaluation. Also, remember that the Quiz will be conducted only in the classroom.

Submission deadline: September 6, 23:59 hrs.

Quiz date: September 7, during class.

Good luck!